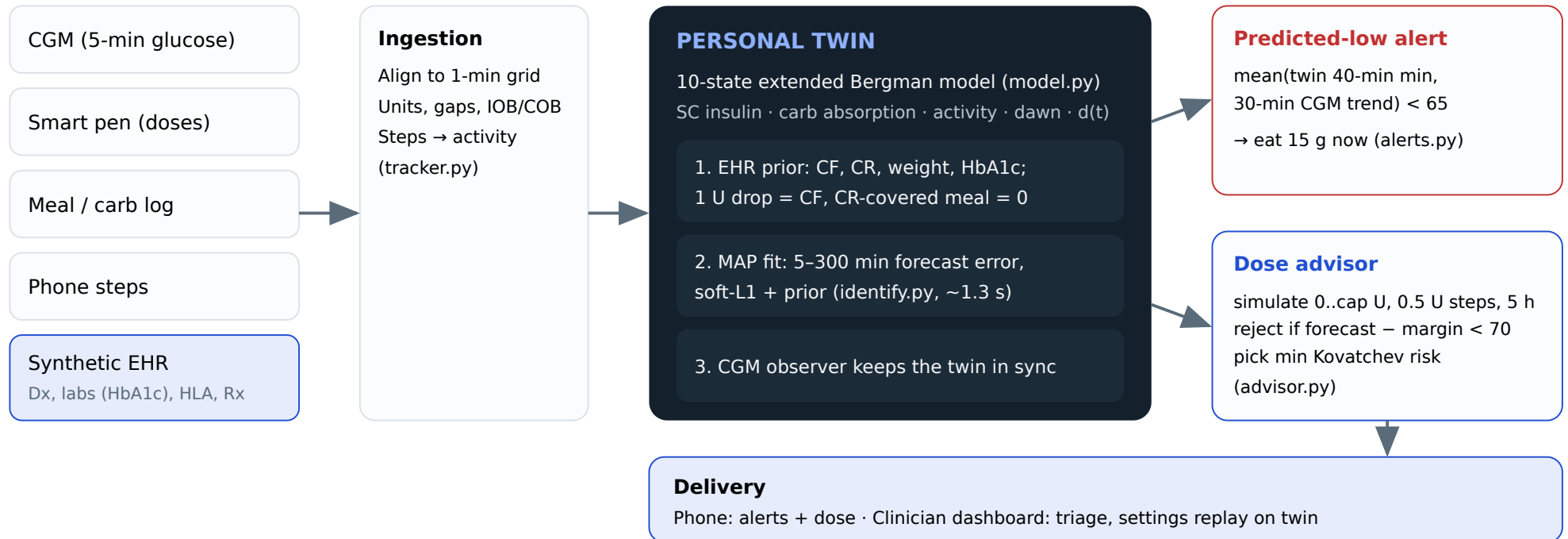


DoseTwin: system architecture

Personal glucose-insulin digital twin for type 1 diabetes on insulin pens. Solid boxes run in deployment; the dashed band is the in-silico evaluation harness.

DATA SOURCES



IN-SILICO EVALUATION HARNESS (not part of deployment)

Hidden truth: UVA/Padova T1D model, 30 virtual patients (uva_padova.py; parity vs simglucose <0.5 mg/dL)

Life simulator: Indian meal timing, carb-count bias + 30% error, missed/late boluses, mis-set clinic settings, walks, CGM noise (cohort.py)

Protocol: dev cohort (seed 7) for tuning + LightGBM baseline; held-out seeds 2026–2028; days 1–7 run-in, days 8–14 paired arms A–E

Outputs: results/study.json, CSVs, figures — bootstrap CIs, Wilcoxon tests (run_study.py, ~15 min CPU)